

BEHOLD!

THE PORTRAITRON 3000

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DESCRIPTION & FUNCTIONALITY

Our latest invention shall capture a person standing in a VERY specific spot! The drawing is executed on paper using our patented MARKERTRON, then struck with a commemorative stamp of OUR OWN FACES, and finally handed over to the subject before they're doomed!

Sketch from Image

- CAPTURE high-definition picture of the subject (3s after keyboard click)
- PROCESS Camera input to strokes
- OPTIMIZES strokes order algorithm

Draw

- PICK UP Marker
- DRAW strokes on page (main goal- our proof of concept)
- LET GO of marker
- PICK UP Stamp

Hand The Drawing (Nice to Have)

- GET the page
- HAND the page to a designated location



HARDWARE & EQUIPMENT

- > UR5e Robotic Arm 🦾
- > 2D High-Def Camera 📷
- > Clipboard 📋
- > Marker 🖋️
- > 3D Printed Marker Mount 🛠️
- > 3D Printed stamp 🖨️
- > Ink 🎨

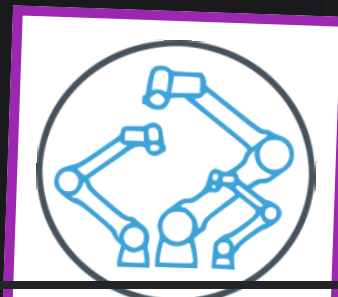
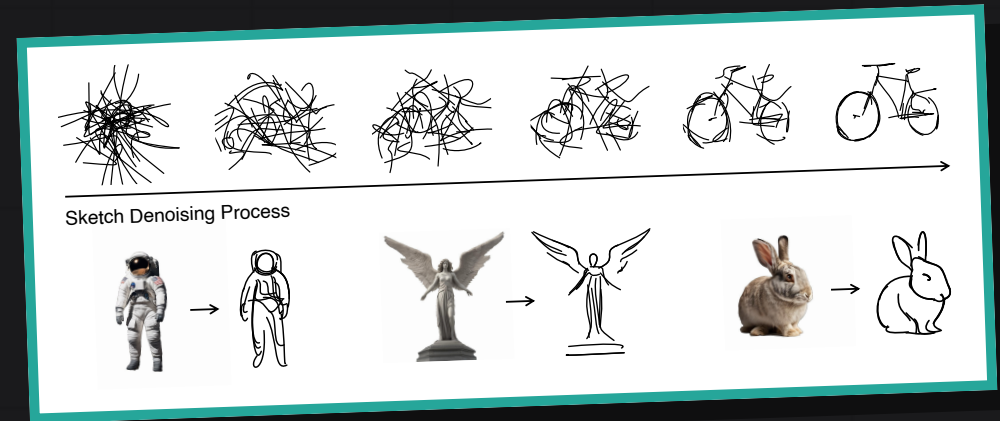


ALGORITHMS

Convert camera input to stroke -
<https://swiftsketch.github.io/>

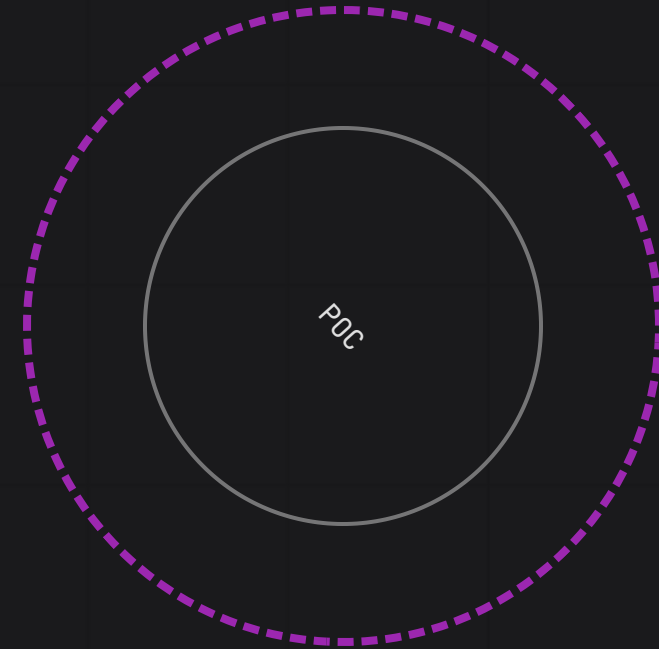
Draw strokes on page - Universal
Robots Real Time Data Exchange (UR-
RTDE)

~~Other Movements (Fixed)~~



PROOF OF CONCEPT

The PORTRAITRON will be able to draw a circle on paper, at this point the vectors are predetermined by a function (rather than by the camera input), and represented in the same format as the output of the image to vector algorithm we intend to use:
<https://swiftsketch.github.io/>



TIMELINE

UP TO POC PRESENTATION:

Design and print marker mount | Implement ability to draw a circle in center of paper (calibration, movement)

AFTER POC PRESENTATION:

Integration of SwiftSketch algorithm | Refinement of drawing ability

CIRCA END OF SEMESTER:

Integration Testing (POTRAITRON is able to convert a picture of a face to vectors and draw it)

UP TO PRESENTATION DAY:

Take picture and send to SwiftSketch algorithm | Design a print stamp | Implement ability to pick up and put down marker mount and stamp | Implement stamping | Implement page



DIVISION OF LABOR

PROOF OF CONCEPT

Avner & Hila

DRAWING REFINEMENT

Hila & Shira

CAMERA & SWIFTSKETCH USE

Avner

FIXED MOVEMENTS

Hila & Shira



THANKS FOR
LISTENING!

TODO

